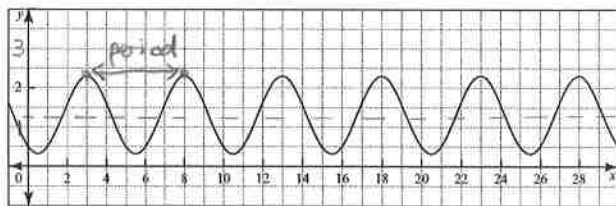


5.5/5.6 Application of Sine and Cosine Functions Worksheet #2

MCR3U

SOLUTIONS

- 1) A motion sensor recorded the motion of a child on a swing. The data was graphed, as shown.



- a) Find the max and min values.

$$\max = 2.25 \quad \min = 0.25$$

- b) Find amplitude $a = \frac{2.25 - 0.25}{2} = 1$

- c) Determine the vertical shift of the function.

$$c = \max - \text{amp} = 2.25 - 1 = 1.25$$

- d) Find the period of the function

$$\text{period} = 8 - 3 = 5$$

- e) Determine the phase shift, if the motion were to be modelled using a sine function.

$$\text{rising midline is } \frac{90}{k} = \frac{90}{72} = 1.25 \text{ to the left of the max.}$$

$$\therefore d = 3 - 1.25 = 1.75$$

- 2) The height of the blade of a wind turbine as it turns through an angle of θ is given by the function $h(\theta) = 8.5 \sin(\theta + 180^\circ) + 40$, with height measured in metres.

- a) Find the maximum and minimum positions of the blade.

$$\begin{aligned} \max &= 40 + 8.5 & \min &= 40 - 8.5 \\ &= 48.5 \text{ m} & &= 31.5 \text{ m} \end{aligned}$$

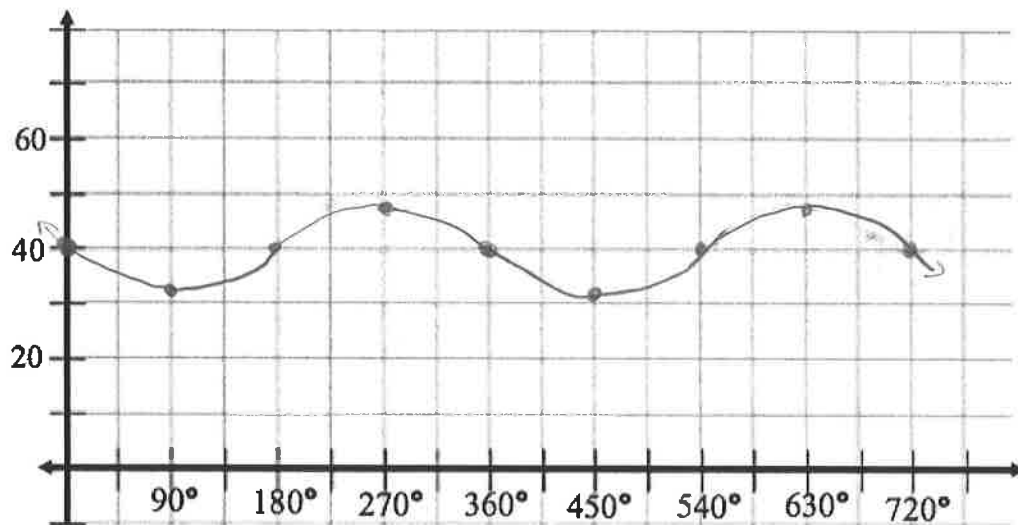
- b) Explain what the value of 40 in the equation represents.

Vertical shift; OR height of the center of the turbine.

- c) Explain what the value of the amplitude represents.

Length of the blade.

d) Sketch the function over two cycles.



3) The height, h , in meters, of the tide in a given location on a given day at t hours after midnight can be modeled using the sinusoidal function $h(t) = 5 \sin[30(t - 5)] + 7$.

a) Find the max and min values for the depth of water.

$$\text{max} = 7 + 5 = 12 \text{ m}$$

$$\text{min} = 7 - 5 = 2 \text{ m}$$

b) What time is high tide? What time is low tide?

rising midline for sine function is moved 5 units right.

$$\text{Max} = 5 + \frac{90}{K} = 5 + \frac{90}{30} = 8 \quad (8:00 \text{ am})$$

$$\text{Min} = 5 - \frac{90}{K} = 5 - \frac{90}{30} = 2 \quad (2:00 \text{ am})$$

Because of the 12 hour period, there is also a max at 8pm and min at 2am.

c) What is the depth of the water at 9:00 am?

$$h(9) = 5 \sin[30(9 - 5)] + 7$$

$$= 5 \sin(120) + 7$$

$$= 11.3 \text{ m}$$

d) Find all the times during a 24-h period when the depth of the water is 3 meters.

$$3 = 5 \sin[30(t - 5)] + 7$$

$$\frac{-4}{5} = \sin[30(t - 5)]$$

$$30(t - 5) = \sin^{-1}\left(\frac{-4}{5}\right)$$

$$t - 5 = \frac{-53.13}{30}$$

$$t - 5 = -1.77$$

$$t = 3.23$$

about 3 hours 14 minutes

This is 1 hour 14 mins after the min; so the height will be the same 1 hour and 14 mins before that; at 12:46 am.

ON BACK →

$h(3)$ at 12:46 am and p.m. (because of 12 hour period)

and

3:14 am and 3:14 p.m.

4) The population, P , of a lakeside town with a large number of seasonal residents can be modeled using the function $P(t) = 5000 \sin[30(t - 7)] + 8000$, where t is the number of months after New Year's Day.

a) Find the max and min values for the population over a whole year.

$$\text{Max} = 8000 + 5000 = 13000$$

$$\text{Min} = 8000 - 5000 = 3000$$

b) When is the population a maximum? When is it a minimum?

max of sine at $\frac{90}{k} = \frac{90}{30} = 3$ but you must shift 7 to the right.

$$\text{min at } -\frac{90}{k} = -\frac{90}{30} = -3$$

so max 10 months after New Years and min 4 months after New Years.

c) What is the population on September 30th?

$$\begin{aligned} P(9) &= 5000 \sin[30(9-7)] + 8000 \\ &= 5000 \sin(60) + 8000 \\ &= 12330 \end{aligned}$$

5) The population of prey in a predator-prey relation is shown. Time is in years since 1985.

a) Determine the max and min values of the population, to the nearest 50. Use these to find the amplitude.

$$\text{Max} = 850$$

$$\text{Min} = 250$$

$$a = \frac{850 - 250}{2}$$

$$a = 300$$

b) Determine the vertical shift, c .

$$c = \text{max} - \text{amp}$$

$$c = 850 - 300$$

$$c = 550$$

c) Determine the phase shift, d .

$$d = 0$$

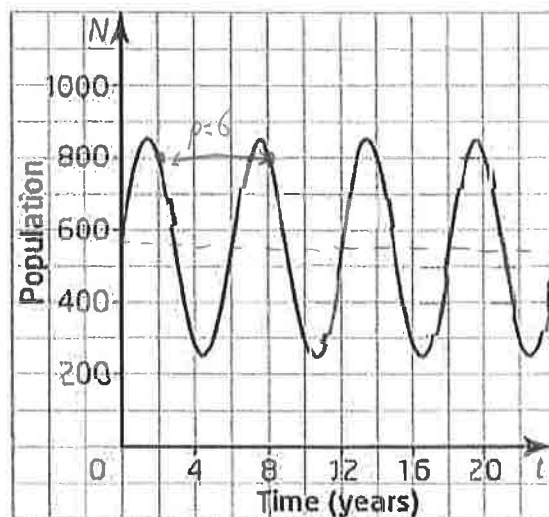
d) Determine the period. Use the period to determine the value of k .

$$\text{Period} = 8 - 2 = 6$$

$$k = \frac{360}{6} = 60$$

e) Model the population versus time with a sinusoidal function.

$$P = 300 \sin(60t) + 550$$



6) The number of millions of visitors that a tourist attraction gets can be modeled using the equation $y = 2.3\sin[30(x + 1)] + 4.1$, where $x = 1$ represents January, $x = 2$ represents February, and so on.

a) Determine the period of the function and explain its meaning.

$$\text{Period} = \frac{360}{30} = 12$$

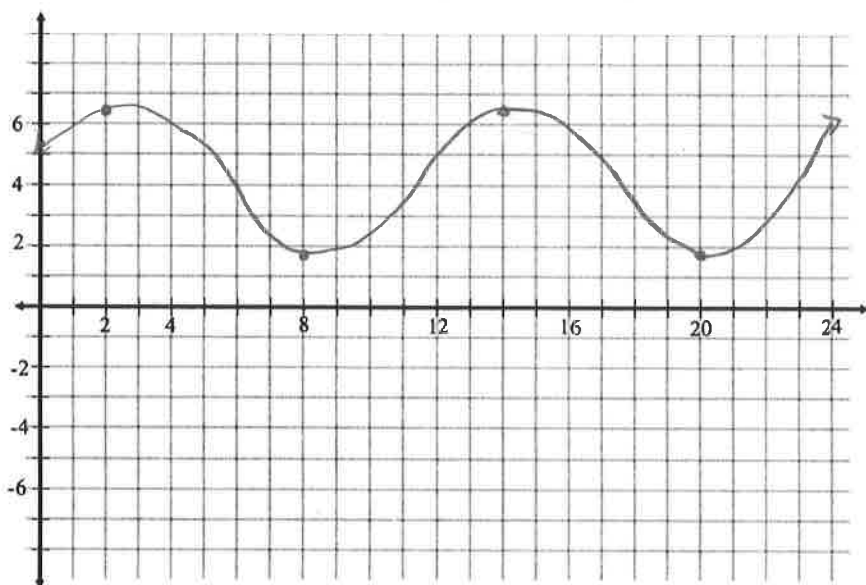
12 months.

b) Graph the function for 12 months. $\text{max} = 4.1 + 2.3 = 6.4$
 $\text{min} = 4.1 - 2.3 = 1.8$

Rising midline at -1

$$\text{First max at } -1 + \frac{90}{30} = -1 + 3 = 2$$

$$\text{First min is } \frac{180}{30} = 6 \text{ after max.}$$



c) Which month has the most visitors?

month 2: February

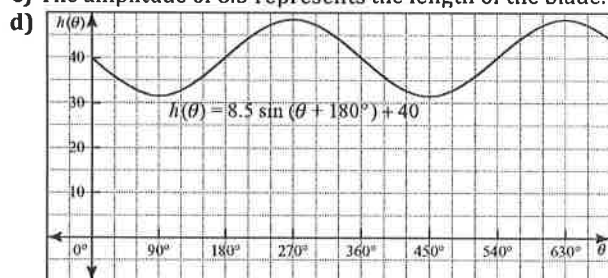
d) Which month has the least visitors?

month 8: August

Answers

- 1) a) maximum 2.25, minimum 0.25
- b) amplitude 1
- c) vertical shift up 1.25
- d) period 5
- e) horizontal shift 1.75 to the right

- 2) a) maximum 48.5, minimum 31.5
- b) The height of the center of the turbine
- c) The amplitude of 8.5 represents the length of the blade.



- 3) a) From the equation, $c = 7$ and $a = 5$, so the function has a midline value of 7 and an amplitude of 5. The maximum height is 12 m and the minimum height is 2 m.

b) From the equation, $k = 30$ and $d = 5$, so the period is 12 h and the phase shift is 5 h right. The first midline value occurs at 5:00 a.m. The first maximum occurs one-quarter period, or 3 h after this, at 8:00 a.m. The previous minimum is 3 h prior to 5:00 a.m., at 2:00 a.m. Because of the 12-h period, there will also be a maximum at 8:00 p.m. and a minimum at 2:00 p.m.

- c) 11.3 m

d) The solution gives a time of approximately 3:14 a.m. This time is 1 h 14 min after the first minimum so the depth should also occur 1 h 14 min before 2:00 a.m., at 12:46 a.m. Because of the 12-h period, the depth will also occur at 12:46 p.m. and 3:14 p.m.

- 4) a) From the equation, $c = 8000$ and $a = 5000$, so the function has a midline value of 8000 and an amplitude of 5000. The maximum population is 13 000 and the minimum population is 3000.

b) From the equation, $k = 30$ and $d = 7$, so the period is 12 months and the phase shift is 7 months right. The initial midline value occurs at $t = 7$. The maximum occurs 3 months later at $t = 10$ (October) and the minimum 3 months earlier at $t = 4$ (April).

- c) 12 330

- 5) a) From the graph the maximum population is approximately 850 animals and the minimum population is approximately 250 animals. The amplitude is approximately 300 animals, so $a = 300$.

b) The vertical shift is the maximum value minus the amplitude, so $c = 550$.

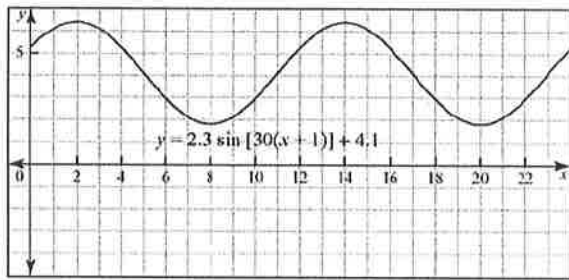
c) The midline intersects the graph at $t = 0$ so no horizontal shift is necessary, so $d = 0$.

d) The pattern repeats every 6 years, so the period is 6 years. $k=60$.

e) A sine function that models the population of prey, N , with respect to time, t , is $N = 300 \sin 60t + 550$.

6) a) 12 months

b)



c) February

d) August

